

Data Science

Syllabus



TECHTONIC-LAB

Tools & Major Topics Covered

- Python Programming
- Advanced Statistics & Mathematics
- SQL
- Advanced Excel
- Power BI / Tableau
- Exploratory Data Analysis (EDA)
- Data Cleaning & Data Wrangling
- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision
- Generative AI & LLMs
- Model Deployment
- Cloud Basics (AWS/Azure)
- Git & GitHub
- End-to-End Industry Projects
- Capstone Project

Introduction to Data Science

- What is Data Science?
 - Data Science Lifecycle
 - Data Scientist Roles & Responsibilities
 - Data Science vs Data Analytics
 - Business Problem Solving with Data
 - Industry Applications of Data Science
 - CRISP-DM Framework
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Python Programming

- Python Fundamentals
- Variables & Data Types
- Lists, Tuples, Dictionaries, Sets
- Conditional Statements
- Loops
- Functions
- Exception Handling
- OOP Concepts
- File Handling
- Modules & Packages

NumPy

- Arrays
- Indexing & Slicing
- Broadcasting
- Mathematical Operations
- Linear Algebra

Pandas

- DataFrames
- Data Cleaning
- Missing Values
- Data Transformation
- GroupBy
- Merge & Join
- Data Manipulation

Data Visualization

- Matplotlib
- Seaborn

- Plotly
 - Statistical Visualizations
 - Interactive Dashboards
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SQL for Data Science

- Database Concepts
 - SQL Queries
 - Joins
 - Subqueries
 - CTEs
 - Window Functions
 - Stored Procedures
 - Views
 - Triggers
 - Data Modeling
 - Normalization
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Advanced Excel

- Lookup Functions
 - Pivot Tables
 - Dashboards
 - Data Validation
 - Power Query
 - Power Pivot
 - VBA & Macros
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Statistics & Mathematics

Statistics

- Descriptive Statistics
- Inferential Statistics
- Probability
- Sampling Techniques
- Hypothesis Testing
- T-Test
- ANOVA
- Chi-Square
- Correlation & Covariance

Mathematics

- Linear Algebra
- Matrices
- Vectors
- Calculus Basics
- Gradient Descent
- Optimization Techniques

Exploratory Data Analysis (EDA)

- Data Profiling
- Data Cleaning
- Feature Understanding
- Outlier Detection
- Missing Value Treatment
- Feature Engineering
- Business Insights Generation

Machine Learning

Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- KNN
- Naive Bayes
- Support Vector Machines (SVM)
- XGBoost

Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- PCA
- Association Rules

Model Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Cross Validation
- Hyperparameter Tuning

Deep Learning

- Introduction to Neural Networks
- Perceptron
- ANN
- TensorFlow

- Keras
 - CNN
 - RNN
 - LSTM
 - Transfer Learning
 - Image Classification
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Natural Language Processing (NLP)

- Text Processing
 - Tokenization
 - Stemming & Lemmatization
 - Sentiment Analysis
 - Text Classification
 - Word Embeddings
 - Transformers
 - BERT Basics
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Generative AI & LLMs

- Introduction to AI & GenAI
 - ChatGPT
 - Prompt Engineering
 - Large Language Models (LLMs)
 - OpenAI APIs
 - LangChain
 - RAG (Retrieval Augmented Generation)
 - AI Agents
 - AI Automation Projects
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Computer Vision

- Image Processing
 - OpenCV Basics
 - Object Detection
 - Face Recognition
 - Image Classification
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Business Intelligence

Power BI / Tableau

- Data Modeling
 - DAX
 - Dashboards
 - Interactive Reports
 - KPI Development
 - Storytelling with Data
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Model Deployment

- Flask / FastAPI
 - Streamlit
 - Model Deployment Basics
 - REST APIs
 - Docker Basics
 - MLOps Introduction
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Cloud Fundamentals (Bonus)

- AWS Basics
- Azure Basics
- Cloud Storage

- ML Services Overview
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Git & GitHub

- Version Control
 - Repositories
 - Branching & Merging
 - Collaboration Workflow
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Projects & Capstone

Guided Projects

- Sales Forecasting
- Customer Churn Prediction
- Credit Risk Analysis
- Recommendation System
- HR Analytics
- Sentiment Analysis
- Image Classification

Capstone Project

- End-to-End Industry Project
- Problem Statement to Deployment
- Dashboard + ML Model + Presentation

Deliverables

- Resume Building
- LinkedIn Optimization
- GitHub Portfolio
- Mock Interviews
- Corporate Grooming
- Placement Assistance

- Industry Capstone Project